

Curriculum Vitae – Camille Godbillot

Postdoctoral fellow

Institut de Recherche pour le Développement (IRD)

camille.godbillot@ird.fr

Research interests: I study the impact of present and future environmental changes on the biomineralization of marine organisms. I have a particular interest in groups that contribute significantly to sedimentary deposits (coccolithophores, diatoms, bivalves, etc.). As contributors to the silica and carbon pumps, their biological production represents a climate feedback that remains poorly constrained and is of interest for future climate projections. Trained in geochemistry, I also develop and use automated imaging and neural network-based counting techniques to study these different groups of marine organisms.

Keywords: Climate change • Nannoplankton • Carbon cycle • Neural networks • Proxy development

1- Professional experience

Full-time research positions

- | | |
|------------------------------|--|
| 01/10/2023 –
(30/09/2025) | Post-doctoral researcher in the DEEP-UP project (as PI) – Institut de Recherche pour le Développement, Aix-en-Provence, France |
| 01/05/2022 –
30/09/2023 | Post-doctoral researcher in the RapMed Project (PI: T. de Garidel-Thoron) – Aix-Marseille Université, Aix-en-Provence, France |

Teaching

- | | |
|----------------------------|--|
| 01/10/2018 –
07/04/2022 | Teaching Assistant at the Earth Sciences Department – Sorbonne Université, Paris, France |
| 01/01/2016-
01/06/2016 | Teaching Assistant at the Languages Department – The University of Chicago, Chicago, USA |

2- Education

- | | |
|----------------------------|---|
| 01/10/2018 –
07/04/2022 | Ph.D. candidate in Earth sciences – Sorbonne Université, Paris, France |
| 01/09/2016 –
30/06/2018 | M.Sc. in Earth Sciences – Sorbonne Université, Paris, France

<i>2nd year dissertation: Culture experiment response of coccolithophore calcification and organic matter production to an Anthropocene-like rise in CO₂ levels.</i>

<i>1st year dissertation: Provenance study of multiple oyster populations using their Rare Earth Element content.</i> |
| 01/09/2013 –
30/06/2016 | B.Sc. in Earth Sciences – Université Pierre et Marie Curie, Paris, France

B.A. in History – Université Paris-Sorbonne, Paris, France

<i>2015-2016: Part of the Student-at-large programme of the University of Chicago, Chicago, USA</i> |

3- Publications

*Peer-reviewed scientific journals (*Supervised student)*

- Godbillot C.**, Pesenti B.*, Leblanc, K., Beaufort, L., Chevalier, C., Di Pane, J., Durrieu De Madron, X., & De Garidel-Thoron, T. (2025). Contrasting Trends in Phytoplankton Diversity, Size Structure, and Carbon Burial Efficiency in the Mediterranean Sea Under Shifting Environmental Conditions. *Journal of Geophysical Research: Oceans*, 130(8), e2025JC022486. <https://doi.org/10.1029/2025JC022486>
- Godbillot, C.**, Marchant, R., Beaufort, L., Leblanc, K., Gally, Y., Le, T. D. Q., Chevalier, C., & De Garidel-Thoron, T. (2024). A New Method for the Detection of Siliceous Microfossils on Sediment Microscope Slides Using Convolutional Neural Networks. *Journal of Geophysical Research: Biogeosciences*, 129(9), e2024JG008047. <https://doi.org/10.1029/2024JG008047>
- Godbillot, C.**, Minoletti, F., Bassinot, F., & Hermoso, M. (2022). Parallel between the isotopic composition of coccolith calcite and carbon levels across Termination II: developing a new paleo-CO₂ probe. *Climate of the Past*, 18(3), 449–464. <https://doi.org/10.5194/cp-18-449-2022>
- Jentzer, M., Agard, P., Bonnet, G., Monié, P., Fournier, M., Whitechurch, H., Omrani, J., Zarrinkoub, M. H., Khatib, M. M., Kohansal, R., Do Couto, D., **Godbillot, C.**, & Ninkabou, D. (2022). The North Sistan orogen (Eastern Iran): Tectono-metamorphic evolution and significance within the Tethyan realm. *Gondwana Research*, 109, 460–492. <https://doi.org/10.1016/j.gr.2022.04.004>
- Mouchi, V., **Godbillot, C.**, Dupont, C., Vella, M., Forest, V., Ulianov, A., Lartaud, F., de Rafélis, M., Emmanuel, L., & Verrecchia, E. P. (2021). Provenance study of oyster shells by LA-ICP-MS. *Journal of Archaeological Science*, 132(May), 105418. <https://doi.org/10.1016/j.jas.2021.105418>
- Hermoso, M., **Godbillot, C.**, & Minoletti, F. (2020). Enhancing Our Palaeoceanographic Toolbox Using Paired Foraminiferal and Coccolith Calcite Measurements From Pelagic Sequences. *Frontiers in Earth Science*, 8(February), 1–5. <https://doi.org/10.3389/feart.2020.00038>
- Mouchi, V., **Godbillot, C.**, Forest, V., Ulianov, A., Lartaud, F., De Rafélis, M., Emmanuel, L., & Verrecchia, E. P. (2020). Rare earth elements in oyster shells: Provenance discrimination and potential vital effects. *Biogeosciences*, 17(8), 2205–2217. <https://doi.org/10.5194/bg-17-2205-2020>

In preparation:

- Walla T., Barras C., Geslin E., **Godbillot C.**, Lanoy L., Marchant R., Dissard, D., & de Garidel-Thoron T. Identification of benthic foraminifera using automated image recognition. In preparation for *Journal of Micropaleontology*.
- Godbillot, C.**, Sanchez Roddriguez, S., Bernales, A., Beaufort, L., Colas., F., De Garidel-Thoron, T., Graco, M. Coccolithophore and diatom dominance and sensitivity to El Nino in the Humboldt system. In preparation for *Biogeosciences*.

Other publications

- De Garidel-Thoron, T., **Godbillot, C.**, Chaabane, S., & Gramoullé, A. (2024). La micropaléontologie à l'ère de l'intelligence artificielle. *Géochronique*, 170.

4- Communications (as main presenter)

*Oral presentations (*Invited)*

- Godbillot C.**, Pesenti B., Leblanc K., Beaufort L., Chevalier C., Di Pane J., Durrieu de Madron X., & de Garidel-Thoron T. Phytoplankton response to environmental shifts in the Mediterranean Sea using novel deep-learning protocols. *EGU General Assembly. Vienna, Austria*. May 2025.
- Godbillot, C.***, Hermoso, M. & Minoletti, F. Reconstruire la concentration en CO₂ atmosphérique dans le passé par l'étude des effets vitaux des coccolithes. *ClimPast 2 – les changements environnementaux pré-Quaternaires. Lyon, France*. September 2023.

Godbillot C., Beaufort L., Leblanc K., Di Pane J., Marchant R., Gally, Y., Chevalier C., D. Q. Le T., Pesenti B., Durrieu de Madron X., & de Garidel-Thoron T. Response of biomineralizing marine phytoplankton to environmental changes in the Mediterranean Sea over the last decade: insights from high-throughput automated workflows. ASLO Aquatic Sciences Meeting 2023. *Palma de Mallorca, Spain*. June 2023.

Godbillot, C., Hermoso, M., & Minoletti, F. Isotopic response of Pleistocene coccoliths to an ambient pCO₂ change: a calibration experiment. 18th International Nannoplankton Association Meeting. *Avignon, France*. September 2022.

Godbillot, C., Hermoso, M., & Minoletti, F. Reconstruction des pCO₂ atmosphériques quaternaires par la géochimie des coccolithes. 27^e édition de la Réunion des Sciences de la Terre, Société géologique de France (SGF). *Lyon, France*. November 2021.

Godbillot, C., Hermoso, M., & Minoletti, F. Probing the use of coccolith vital effects as a proxy for past CO₂ concentrations – Insights from Termination II in the Northern Atlantic Ocean. 31st Goldschmidt Conference, *Lyon, France (virtual meeting)*. July 2021.

Poster communications

Godbillot, C., de Garidel-Thoron, T., Beaufort., L., Leblanc, K., Chevalier, C., D.Q. Le, T., Gally., Y., & Durrieu de Madron., X. Response of marine nannoplankton to environmental forcings: Insights from automated imaging methods on community shifts in the Gulf of Lions over the last decade. 18th International Nannoplankton Association Meeting. *Avignon, France*. September 2022.

5- Research projects and funding

2024-2025 – CEREGE Internal call for projects – Microfluidics and Micromanipulation (MiMi) – 6000 euros

2023-2024 – CEREGE Internal call for projects – Financial support for a 2-month visit at IMARPE (Peru) – 6500 euros

2023-2025 – Postdoctoral fellowship – Competitive 2-year fellowship awarded to the DEEP-UP project (Deep learning applied to Phytoplankton from the Peruvian Upwelling) by the Institute of Research for Development (IRD).

2018-2021 – Ph.D. Fellowship – Competitive 3-year fellowship awarded by the “Géosciences, Ressources Naturelles et Environnement” (GRNE) Doctoral School of Sorbonne Université.

6- Awards

Recipient of the City of Marseille's grant for young researchers (2022)

7- Mentoring

01/03/ 2025 **Primary supervisor of Guilhem Le Bars (Master's 1)**

30/06/2025 Project: "Changes in the phytoplankton community during El Niño events"

01/01/2024 – **Co-supervisor of Mathias Ageron (Master's 2, supervisor T. de Garidel-Thoron)**

30/06/2024 Project: "Variability of faunas and morphometric traits of benthic and planktonic foraminifera in the eastern Mediterranean during the sapropel S1 episode: development of an automation method" – training in the development and analysis of automated detection models using convolutional neural networks.

01/01/2023 – **Primary supervisor of Baptiste Pesenti (Master's 2)**

30/06/2023 Project: "Study of community changes in calcareous and siliceous phytoplankton from the DYFAMED sediment trap series in the Ligurian Sea" – training in sample preparation, automated microscopy techniques, data processing, and visualization (R, etc.).

8- Skills

Micropaleontology: Taxonomy of coccoliths and diatoms, sample preparation from marine bio-carbonates for chemical analyses and automated imaging.

Deep learning: Development of an image processing workflow (annotation, detection, classification) for various phytoplankton groups.

Geochemical analyses: Stable isotope chemistry ($\delta^{18}\text{O}$, $\delta^{13}\text{C}$) of carbonates, Inorganic geochemistry of Major, Trace and Rare Earth Elements using LA-ICP-MS, XRF, Optical and SEM cathodoluminescence, EPMA, and CHNS combustion analyses

Cell culture experiments: Coccolithophore culturing techniques including medium preparation, cell acclimation and monitoring of population growth rates and sizes.

GIS: ArcGis, Ocean Data View

IT: Data, spatial information, and timeseries analyses using R, Python, and Microsoft Excel

Languages: Fluent French and English (TOEFL iBT 117/120 in 2018); Basic Spanish

9- Professional and Academic Services

Referee

Geochimica et Cosmochimica Acta (Elsevier), *AI in Geosciences* (Elsevier), *Scientific Reports* (Springer Nature), *Geo-Marine Letters* (Springer Nature).

Session Chair

Co-Convener for the “Evolution and Ecology” session at the 18th International Nannoplankton Association Meeting, Avignon, France (September 2022).

Workshop participation:

“Reconnaissance et Automatisation par IA”, Angers, France (December 2024).

“L’utilisation de l’IA pour analyse de données issues de séries longues”, Paris, France (June 2024).